

General Physics - I.

Motion in 1D. (high school physics + calculus)

Typical approach in theoretical physics:



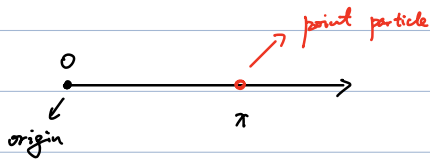
e.g. magnets $\rightarrow$ approximate electrons as arrows (Good enough for most magnets)

$\downarrow$ when not good enough (SC, QSL, ...)

arrow + quantum mechanics $\Rightarrow$ quantum spin

Review of high-school physics (back to class)

Coordinate system for 1D:



position:

π

displacement:

$$\Delta \pi = \pi_2 - \pi_1$$

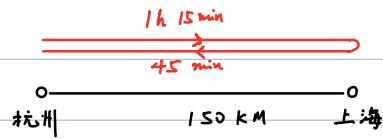
average velocity (速度):

$$v_{\text{avg}} = \frac{\pi_2 - \pi_1}{t_2 - t_1}$$

average speed (速率):

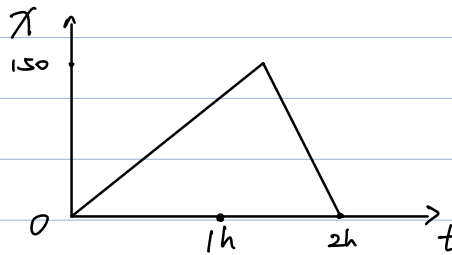
$$s_{\text{avg}} = \frac{\text{total distance}}{t_2 - t_1}$$

Example.

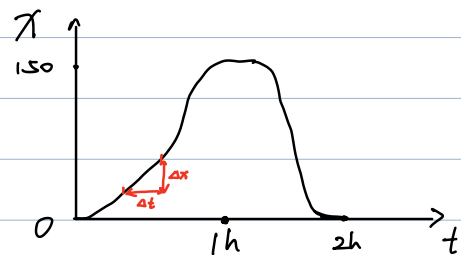


$$V_{avg} = 0$$

$$S_{avg} = \frac{300 \text{ km}}{2 \text{ h}}$$



reality
→



Instantaneous velocity:

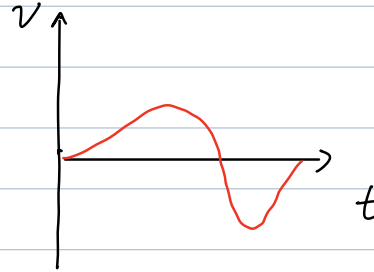
$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \quad ("-" \text{ ok})$$

Instantaneous speed:

$$|v| \geq 0$$



reality
→



Example.

$$x = At^n$$

velocity: $v = \frac{dx}{dt} = An t^{n-1}$

(similarly. we can define)

Average acceleration :

$$a_{\text{avg}} = \frac{v_2 - v_1}{t_2 - t_1}$$

↓↓

Instantaneous acceleration :

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

Example.

$$v = 40 - 5t^2 \quad \text{m/s,} \quad (\text{unit of } t: \text{s})$$

$$\Rightarrow a = \frac{dv}{dt} = -10t \quad \text{m/s}^2$$

@ $t = 10\text{s}$,

$$a = -100 \quad \text{m/s}^2 \quad (\text{note: } < 0)$$

Units

Length (L): m, km, mm, ...

Mass (M): kg, g, mg, ...

Time (T): s, ns, ...

SI (International system of Units): m, kg, s, ...

Dimension:

$$[v] = L/T$$

Integration.

$$(\text{just barnt}) \quad \begin{cases} v = \frac{dx}{dt} \\ a = \frac{dv}{dt} \end{cases}$$

For given $a(t)$:

$$dv = a(t) dt$$

$$\Rightarrow v(t) = \int_{t_i}^t a(t') dt' + C,$$

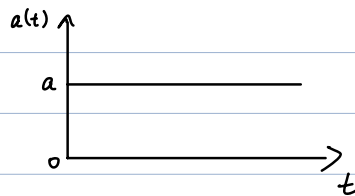
$$v(t_i) \equiv v_i \Rightarrow C_i = v_i \Rightarrow v(t) = \int_{t_i}^t a(t') dt' + v_i \quad (\text{"i" for initial}) \quad (3)$$

Similarly, for given $v(t)$:

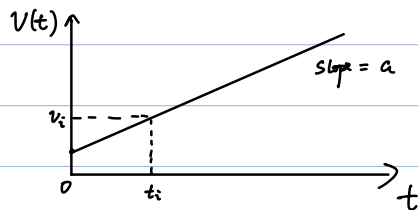
$$\begin{aligned} x(t) &= \int_{t_i}^t v(t') dt' + x_i \\ &= \int_{t_i}^t dt' \left[\int_{t_i}^{t'} a(t'') dt'' + v_i \right] + x_i \\ &= x_i + (t - t_i) v_i + \int_{t_i}^t dt' \int_{t_i}^{t'} a(t'') dt'' \end{aligned}$$

Example. Constant acceleration.

$$a(t) \equiv a \quad (\text{time-independent})$$



Apply
$$\begin{aligned} v(t) &= \int_{t_i}^t a(t') dt' + v_i \\ &= (t - t_i) a + v_i \end{aligned}$$



set $t_i = 0$, $v_i = v_0$,

$$\Rightarrow \boxed{v(t) = v_0 + at}$$

Similarly, $t_i = 0$, $x_i = x_0$, $v_i = v_0$, $a(t) = a$

$$\begin{aligned} x(t) &= x_i + (t - t_i) v_i + \int_{t_i}^t dt' \int_{t_i}^{t'} a(t'') dt'' \\ &= x_0 + v_0 t + a \int_0^t dt' (t' - 0) \\ &= \boxed{x_0 + v_0 t + \frac{1}{2} a t^2} \end{aligned}$$

Other forms:

$$\begin{aligned} x(t) - x_0 &= \frac{1}{2} v_0 t + \frac{t}{2} (v_0 + at) \\ &= \boxed{\frac{1}{2} (v_0 + v_f) t} \\ &= \frac{1}{2} (v_0 + v_f) \frac{v_f - v_0}{a} \\ &= \boxed{\frac{v_f^2 - v_0^2}{2a}} \end{aligned}$$

Exercise on dimension analysis

$$[x(t) - x_0] = \left[\frac{1}{2} (v_0 + v_f) t \right] = L$$

$$\Rightarrow [v][t] = L$$

$$\Rightarrow [v] = L/T$$

Exercise for non-constant acceleration

$$a(t) = a t^\alpha \quad (\alpha = 0 \Rightarrow \text{constant})$$

suppose $\alpha > 0$:

$$v(t) = \int_{t_i}^t a(t') dt' + v_i$$

$$= a \int_{t_i}^t (t')^\alpha dt' + v_i$$

$$= \frac{a}{\alpha+1} (t')^{\alpha+1} \Big|_{t_i}^t + v_i$$

$$= \frac{a}{\alpha+1} (t^{\alpha+1} - t_i^{\alpha+1}) + v_i$$

$$x(t) = \dots$$

• Significant figures

$$7.62 \times 10^3 \text{ m} \quad \rightarrow 3 \text{ sig figures}$$

$\downarrow$
uncertain

— Leading zeros : NOT counted

$$0.762 \times 10^4 \text{ m} \quad \rightarrow 3 \text{ sig figs}$$

— Trailing zeros : Counted

$$7.6200 \times 10^3 \text{ m} \quad \rightarrow 5 \text{ sig figs}$$

— Addition / Subtraction : first uncertain digit

$$\begin{array}{r} 213.25 \\ 16.7 \\ + 0.124 \\ \hline 230.074 \end{array} \quad \rightarrow 230.1 \quad 4 \text{ sig figs}$$

— Multiplication / Division : least sig fig

$$325.78 \times 0.0145 \div 789.2 = 0.00599 \quad 3 \text{ sig figs}$$

— More delicate rules : should be covered in experiment classes ...

⑤

• Uncertainty propagation

$$y = y(x_1, x_2, \dots, x_n)$$

$$\Delta y = \sqrt{\left(\frac{\partial y}{\partial x_1}\right)^2 (\Delta x_1)^2 + \dots + \left(\frac{\partial y}{\partial x_n}\right)^2 (\Delta x_n)^2}$$

Example

$$y = x_1 + x_2$$

$$\Delta y = \sqrt{(\Delta x_1)^2 + (\Delta x_2)^2}$$

Example

$$y = x_1 x_2$$

$$\Delta y = \sqrt{(x_2 \Delta x_1)^2 + (x_1 \Delta x_2)^2}$$

$$\Rightarrow \frac{\Delta y}{|y|} = \sqrt{\left(\frac{\Delta x_1}{x_1}\right)^2 + \left(\frac{\Delta x_2}{x_2}\right)^2}$$

Example

$$y = x_1 / x_2$$

$$\Delta y = \sqrt{\left(\frac{1}{x_2} \Delta x_1\right)^2 + \left(\frac{x_1}{x_2^2} \Delta x_2\right)^2}$$

$$\Rightarrow \frac{\Delta y}{|y|} = \sqrt{\left(\frac{\Delta x_1}{x_1}\right)^2 + \left(\frac{\Delta x_2}{x_2}\right)^2}$$

$$\overset{x_1}{\uparrow} \quad \overset{\Delta x_1}{\uparrow} \quad \quad \overset{x_2}{\uparrow} \quad \overset{\Delta x_2}{\uparrow}$$

$$(5.5 \pm 0.1) \text{ cm} \times (6.4 \pm 0.1) \text{ cm}$$

$$= 35 \pm (35 \times 0.024) \text{ cm}^2$$

$$= 35 \pm 1 \text{ cm}^2$$

$$\sqrt{\left(\frac{0.1}{5.5}\right)^2 + \left(\frac{0.1}{6.4}\right)^2} \approx 0.024$$

$$35 \times 0.024 \approx 0.84$$